

May-22 - Probability

1. Random Experiment 2. Trial 3. Outcome 4. Sample Space 5. Event {Set of outcomes}

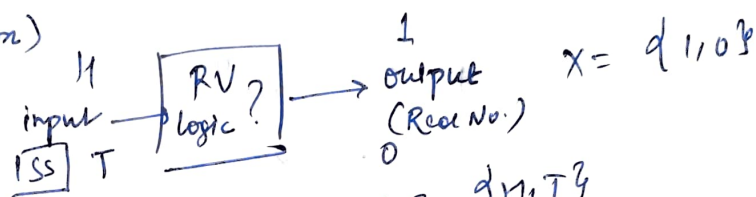
Types of Events

1. Simple Event 2. Compound Event 3. Independent 4. Dependent
5. Mutually Exclusive 6. Exhaustive 7. Impossible & Certain
At least 1 will occur

Empirical Probability vs Theoretical P.
 $n/T \rightarrow \frac{55}{100}$ $n/T : (\frac{1}{2})$

$$3C_2 \times \left(\frac{1}{2}\right)^2 \times \left(\frac{1}{2}\right)^1 = \frac{3}{8}$$

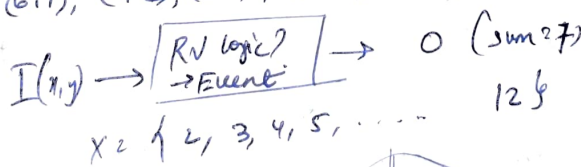
Random Variable (is a function)
↳ misleading.



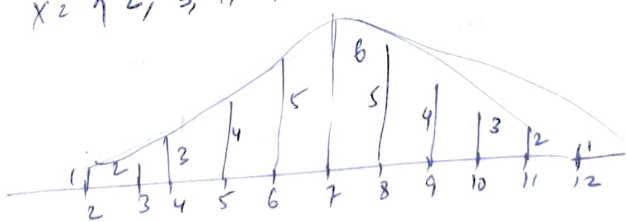
R. Exp. $\rightarrow$ Tossing a coin $\{H, T\}$
Rolling a die $\rightarrow \{1, 2, 3, 4, 5, 6\}$

Transformation $1/0 \rightarrow$ how we choose to define a random variable.
How to define R.V often depends on the specific aspects that you're interested in studying.

Rolling 2 die $\rightarrow$ Getting sum of 7
RV depends on the event
(1,6), (6,1), (2,5), (5,2), (3,4), (4,3)



2	3	4	5	6	7
3	4	5	6	7	8
4	5	6	7	8	9
5	6	7	8	9	10
6	7	8	9	10	11
7	8	9	10	11	12



RV $\rightarrow$ i) Continuous
ii) Discrete

X	1	0
p(x)	1/2	1/2

X	2	3	4	5	6	7	8	9	10	11	12
p(x)	1/36	2/36	3/36	4/36	5/36	6/36	5/36	4/36	3/36	2/36	1/36

probability distr. of random variable

when x is very high
the $p(x) = ?$ is difficult to find

$P(x) = \frac{f(x)}{L}$, mathematical relation (Probability distribution function, PDF)

PMF for discrete case.

Mean of a Random Variable

x	1	2	3	4	5	6
$P(x)$	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$

1000 trials

$$\bar{X} = \frac{1+2+\dots+4+1+\dots+6}{1000}$$

$$5 \quad 3 \quad 4 \quad 5 \quad 3 \quad 3 \quad , \quad \left[\frac{1}{3}(5) + \frac{1}{6}(4) + \frac{1}{2}(3) \right] \rightarrow \text{mean}$$

$$P(5) \times 5 + P(4) \times 4 + P(3) \times 3$$

1, 2, 3, 4, 5, 6

$$P(1-6) \frac{21}{6} = 3.5$$

The mean of a random variable, is often called the expected value is essentially the avg. outcome of a random process that is repeated many times. Weighted - avg of possible outcomes of R.V. each outcome is weighted by probability of occurrence.

$$E[X] = \sum_{i=1}^n x_i P(x_i)$$

Variance of R.V

x	1	2	3	4	5	6
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$$E[X] = \frac{21}{6} = 3.5$$

$$\text{Normal var} = \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n-1}$$

$$\text{var}(X) = \frac{E[(X_i - E[X])^2]}{n}$$

$$= \frac{(1-3.5)^2}{6} + \frac{(2-3.5)^2}{6} + \frac{(3-3.5)^2}{6} + \frac{(4-3.5)^2}{6} + \frac{(5-3.5)^2}{6} + \frac{(6-3.5)^2}{6}$$

$$\text{var}(X) = E[(X - E(X))^2]$$

$$= \frac{1}{6}(2.5^2 + 1.5^2 + 0.5^2 + 0.5^2 + 2.5^2 + 2.5^2) = 2.916$$

$$\text{var}(x) = E[X^2 + E(x)^2 - 2 \cdot x \cdot E(x)]$$

$$= E[X^2] + E[E(x)^2] - 2 \cdot E(x) \cdot \underset{\substack{\downarrow \\ E(x)}}{E[E(x)]}$$

$$= \cancel{E[E(x)^2]} - \cancel{E[x]^2}$$

$$E[X^2] - (E[X])^2$$

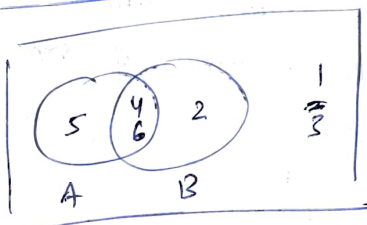
$$\text{var}(x) = E[X^2] - (E[X])^2$$

$$15.166 - 3.5^2 = 2.916$$

$$\frac{1+4+9+16+25+36}{6} = 15.166$$

Probability Vid-2

A → num (4, 5, 6)
B → even num (2, 4, 6)



$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{3}{6} + \frac{3}{6} - \frac{2}{6} = \frac{4}{6} - \frac{2}{6} = \frac{2}{6} = \frac{1}{3}$$

$$P(A \cup B) = \frac{2}{6} = \frac{1}{3}$$

Contingency Tables

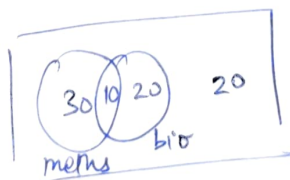
A → 4 (4, 5, 6) 4
B → even no. (2, 4, 6) 10

	even	not even
≥ 4	2	1
≤ 4	1	2

100 Students

40 - only bio
30 - only maths
10 - both

	math	no math
bio	10	40
not-bio	30	20



Joint Probability

	①	2	3
①	0.089	0.108	0.412
1	0.1526	0.091	0.133

when we calculate probability of two events is called joint probability

Marginal Probability / Simple / Unconditional probability P(A)

X	Pass	1	2	3	All
Surviving	0	80	92	372	544 → 0.616 → died
	1	136	87	119	342 → 0.383 → survived
All		216	184	491	891 →
		0.24	0.20	0.55	
		Pass 1	Pass 2	Pass 3	

irrespective of the outcome of some other event.

$$\begin{aligned} P(Y=0) &= 0.61 \\ P(Y=1) &= 0.39 \end{aligned}$$

↓
marginal probability dist. of Y

→ marginal probability dist. of X

$$P(X=1,2,3)$$

In pass, pd. crosses, margin = 'True'

Conditional Probability → P(Event given another event has occurred), P(A/B)

$$P(\text{At least 2 heads} \mid \text{At least 1 coin shows heads})$$

$$P(\text{sum} = 7 \mid \text{odd no.})$$

$$\frac{3}{13}$$

$$= \frac{3}{13}$$

$$= \frac{3}{6} \times \frac{3}{6} \times \frac{3}{6}$$

$$\frac{3}{216}$$

$$A \rightarrow D_1 = 2 \rightarrow 3$$

$$B \rightarrow D_1 + D_2 \leq 5 \rightarrow 10$$

$$P(A/B) = \frac{3}{10}$$

$$\frac{3}{10}$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \rightarrow \text{joint probability} \quad P(B) \rightarrow \text{marginal probability of } B$$

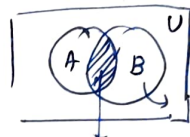
$$p(y=0|x=3) = \frac{p(y=0 \cap x=3)}{p(x=3)} = \frac{0.417}{0.54} = 0.75$$

$$p(y=0|x=2) = 0.52$$

normalize = All
Joint P → for all

	'Row/index'	'Columns'
	$P(X Y)$	$P(Y X)$

Intuition behind $\rightarrow P(A|B) = \frac{P(A \cap B)}{P(B)}$



$$P(B) = B/U$$

$$P(A \cap B) = A \cap B / U$$

$$\Rightarrow P(A|B) = \frac{P(A \cap B)}{P(B)}$$

~~Intuition~~ Independence vs (Mutually Exclusive) $\rightarrow$ Can't both occur at same time

Statistically independent	Mutually Exclusive	Dependent
$P(A B) = P(A)$	0	$\frac{P(A \cap B)}{P(B)}$
$P(B A) = P(B)$	0	$\frac{P(A \cap B)}{P(A)}$
$P(A \cap B) = P(A) \cdot P(B)$	0	$P(A B)P(B) \text{ or } P(B A)P(A)$

$$P(A \cap B) = P(A) \cdot P(B)$$

for independent events.

Bayes Theorem $\rightarrow$ describes how to update the probabilities of hypothesis when given evidence. Variety of fields $\rightarrow$ ML, Statistics, game theory.

$$\boxed{P(A|B)} = \underbrace{\boxed{P(A)}}_{\text{prior}} \times \frac{\boxed{P(B|A)}}{\boxed{P(B)}} \rightarrow \text{likelihood} \quad \rightarrow \text{marginal evidence.}$$

$$P(A|B)P(B) = P(B|A)P(A)$$

gender survival

M $\rightarrow$ 0
M $\rightarrow$ 0
F $\rightarrow$ 1
F $\rightarrow$ 0
M $\rightarrow$ 1

$\downarrow$ M $\rightarrow$ 0, 1

$$P(0|M) = \frac{2}{3}$$

$$P(0|M) = P(0) \times \frac{P(M|0)}{P(M)} = \frac{3}{5} \times \frac{2/3}{3/5} = \frac{2}{3}$$